Maharaja Agrasen Institute of Technology

Department of Information Technology

COLLEGE ALUMNI INTERACTION SYSTEM

Mini Project Report

1. Title

College Alumni Interaction System using MySQL and JDBC

2. Objective

The objective of this project is to design and implement a database-driven application that facilitates interaction between students and alumni. The system focuses on **profile evaluation and privacy**, allowing users to register, login, and manage their profiles securely.

3. Introduction

In modern educational institutions, maintaining a strong alumni network is essential for career guidance, mentorship, and professional growth. This project aims to develop a system where students and alumni can connect, share information, and evaluate profiles while maintaining privacy controls.

The system is built using:

- **MySQL** for database management
- **Java (JDBC)** for backend connectivity

4. Problem Statement

Traditional alumni systems lack:

- Secure login mechanisms
- Structured data storage
- Privacy control over user information
- Profile evaluation features

This project solves these issues by providing a **secure, scalable, and interactive platform**.

5. System Requirements

Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM

Software Requirements

- MySQL Server
- Java (JDK 8 or above)
- MySQL Connector/J (JDBC Driver)
- VS Code / Any Java IDE

6. Database Design

Main Tables

1. **Users**
   - Stores user details like name, email, role, company
   - Includes privacy settings
2. **AlumniProfiles**
   - Stores alumni-specific information
3. **StudentProfiles**
   - Stores student-specific information
4. **ProfileRatings**
   - Used for profile evaluation
5. **Endorsements**
   - Stores skill endorsements
6. **Connections**
   - Manages user networking
7. **Messages**
   - Enables communication
8. **PrivacySettings**
   - Controls access to user data

7. Features of the System

- User Registration
- User Login Authentication
- Profile Viewing
- Profile Update
- Account Deletion

- Profile Evaluation (Ratings)
- Privacy Control

8. Methodology

The system follows these steps:

1. **Database Creation**
    - Tables created using SQL queries
    - Relationships defined using foreign keys
2. **Backend Development**
    - JDBC used to connect Java with MySQL
    - SQL queries executed using PreparedStatement
3. **User Interaction**
    - Console-based interface
    - Menu-driven operations

9. Implementation

JDBC Connection

The application connects to the database using:

Connection conn = DriverManager.getConnection(URL, USER, PASSWORD);

Operations Implemented

- INSERT → Registration
- SELECT → Login & View Profile
- UPDATE → Modify Profile
- DELETE → Remove Account

10. Output

Sample Output

- Registration Successful
- Login Successful
- Profile Details Displayed
- Profile Updated
- Account Deleted

11. Advantages

- Simple and user-friendly
- Secure login system
- Efficient data management
- Privacy-focused design
- Scalable for future improvements

12. Limitations

- Console-based interface (no GUI)
- Password stored without encryption (basic version)
- Limited real-time interaction

13. Future Scope

- Add GUI using Java Swing or JavaFX
- Implement password encryption
- Develop web-based version
- Add real-time messaging system
- Include admin panel

14. Conclusion

The College Alumni Interaction System successfully demonstrates how a database can be integrated with a Java application using JDBC. It provides essential features such as user authentication, profile management, and privacy control. The project can be further enhanced into a full-scale application.

15. References

- MySQL Documentation
- Java JDBC Documentation
- Oracle Official Website

16. Student Details

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